

Code-Layout ,Readability and Reusability

Date	6July2026
Team ID	
Project Name	Plugging into the Future : An Exploration of Electricity Consumption Patterns
Maximum Marks	5Marks

Code-Layout , Readability and Reusability:

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes/No/ Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation are maintained through out the source code.	Yes	Improves read ability and code consistency
2	Proper File Structure	Project files are organized into folders such as templates, static, models, datasets, and application files	Yes	Makes the project easy to maintain and navigate
3	Meaningful Variable Names	Variables are named clearly to represent their purpose and functionality	Yes	Enhances code read ability and understanding
4	Function/Method Names	Functions are named according to the operations they perform	Yes	Improves maintain ability and code clarity
5	Code Comments	Important sections of the code include comments explaining their functionality	Yes	Helps developers understand and maintain the code
6	Modular Design	Code is divided into reusable modules for preprocessing, model training , prediction , and Web application	Yes	Promotes code reusability and easier updates
7	No Redundant Code	Duplicate and unnecessary code has been removed	Yes	Reduces complexity and improves efficiency
8	Error Handling	Input validation and exception handling	Yes	Ensures reliable and robust application execution

Reusable Components / Modules:

S.No	Component/Module Name	Language/ Technology	Where Reused	Reusability Level (High/Medium/Low)
1	Data Preprocessing Module	Python , Pandas	Model Training & Prediction	High
2	Machine Learning Model	Scikit -learn	Prediction Module	High
3	Flask Routes	Flask(Python)	Web Application	High
4	HTML Templates	HTML,CSS	User Interface	Medium
5	Prediction Function	Python	Flask Backend	High

Overall Code Quality Assessment:

Aspect	Rating(1-5)	Comments
Code Layout & Structure	5	Well – organized project structure
Readability	5	Easy – to – read code with meaningful names
Reusability	5	Functions and modules can be reused easily
Documentation/ Comments	4	Code is documented with necessary comments
Overall Score	4.8/5	Clean , maintainable, and reusable code base